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Tutoring Section 2 Class 3

Numerical Problems in Machining

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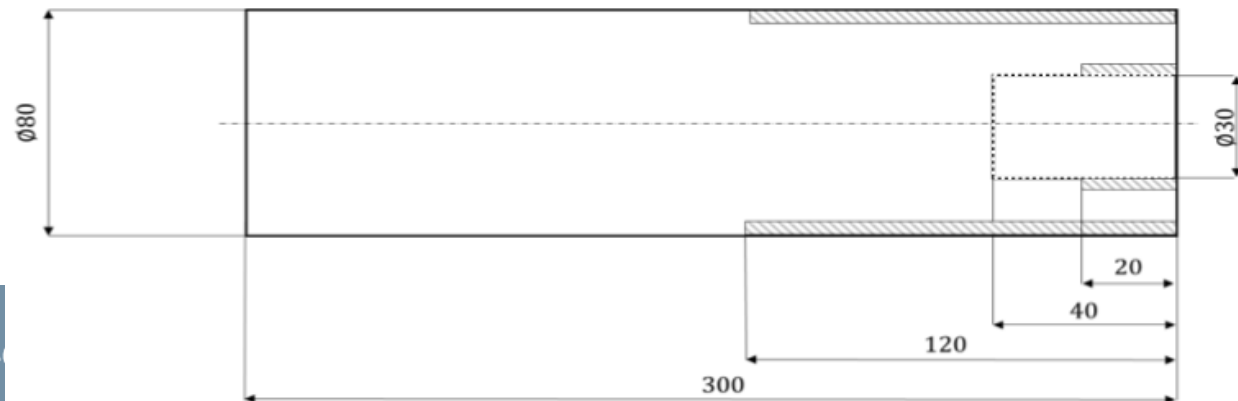
17:30 – 19:00 , 6/Dec/2022
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MACHINING : TURNING OPERATIONS

■ Problem 5 (12/07/2021 exam)

The mechanical component in the figure must be machined on a lathe. The machining operations are an external cylindrical turning of a length equal to 120 mm and an internal cylindrical turning of a length equal to 20 mm. A metal bar of 80 mm diameter is mounted on a self-centering 3-jaw chuck and machined according to the following parameters (the two machining operations are carried out with tools with identical characteristics):

- External cylindrical turning: depth of cut = 1 mm, feed = 0,4 mm/rev.
 - Internal cylindrical turning: depth of cut = 0,5 mm, feed = 0,1 mm/rev.
 - Other data: Cutting pressure: $k_{c0,4} = 2600$ MPa, Kronenberg coefficient $x = 0,29$, Main entering angle = 95° , Secondary entering angle = 10° , Tool nose radius = 0,4 mm, Nominal lathe power = 5 kW, Lathe efficiency = 90%.
1. Determine the cutting force, the cutting power, and the cutting time for the external cylindrical turning considering a cutting speed is 250 m/min and determine if the machining operation is feasible.
 2. Determine the average roughness on the machined surface of the hole.



MILLING & DRILLING OPERATIONS

■ Problem 1 (21/Jun/2021 exam)

Consider a machining cycle to obtain a through hole with a diameter of 40 mm, a depth of 80 mm and a tolerance grade equal to IT7. The machining cycle involves a drilling operation with a twist drill bit with a diameter equal to 37 mm and a reaming operation. The machining operations are carried out on a machine tool with a nominal power equal to 15 kW and an efficiency equal to 90%. Determine:

- the cutting torque for the reaming operation;
- the cutting power for the reaming operation;
- the machining time for the reaming operation;
- Is the reaming operation feasible on the selected machine tool?

Reaming data:

- Cutting speed: $v_c = 120$ m/min
- Feed: $f = 1$ mm/revolution
- Number of cutting edges: $Z = 8$
- Cutting pressure: $k_c = 2100$ MPa
- Extra path in and out: 2 mm each

